Nabhay Khanna

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Education

University of California, San Diego — San Diego, CA
B.S. in Computer Science — *Expected June 2030*
- **CSES (Computer Science and Engineering Society)**, Member — Oct 2026 – Present
  * Working on the Press Club Project, a year-long team project involving scraping news websites

Delhi Public School R.K. Puram — New Delhi, India
CBSE Class XII (94.6%) — *2026*
- **AP Courses:** Calculus BC, Physics C: Mechanics, Computer Science A, Computer Science Principles

Experience

DocsDocs (Dental Practice Management System) — Apr 2026 – Jun 2026
ML Engineer
- Built a computer vision pipeline to detect and segment defects in dental X-rays, training Mask R-CNN on a corpus merged from 4 open-source datasets
- Reached a Dice score of 0.84 on held-out data by standardizing heterogeneous annotation formats into a single training and evaluation pipeline
- Integrated Sikka's API to automate insurance and patient data exchange with clearinghouses

Coherent Education (AI Learning Platform for Schools) — May 2025 – Jan 2026
CTO
- Raised diagram-generation accuracy from 30% (base LLMs) to 75% using structured generation and subject-specific reasoning pipelines
- Designed RAG pipelines for question answering and cut latency and token cost by routing tasks across multiple models by requirement
- Secured a $10,000 pilot agreement with the Ladli Foundation to deploy the platform in 100 school computer labs

Host4Asia (Web Development Agency) — Apr 2024 – Jun 2024
Software Developer
- Developed the Python/SQLite backend for the Indian Fertility Society website, with database-backed APIs and validation logic that let staff update content without frontend changes and reduced content errors

Projects

MyEye | *Python, CRAFT, Gemini, pyttsx3, Raspberry Pi 5* — Top 2 in India, IRIS National Science Fair
- Created an assistive system that reads handwritten text aloud using CRAFT text detection with a Gemini correction pass, plus a voice assistant that describes the user's surroundings
- Optimized the full pipeline to run entirely on a single Raspberry Pi 5

Custom Load Balancer | *Go, Python* — github.com/Nabhay/LoadBalancer | loadbalancer.khanna.app
- Engineered a Go load balancer in front of a Python backend, sustaining 6M+ requests in 3 hours (∼550 req/s) on a single vCPU
- Mitigated DDoS traffic with request validation and a leaky-bucket rate limiter; cached game levels and leaderboards for 200 concurrent players

Honors & Competitions

- **Tetr Disrupt:** 1st place in India (40 teams)
- **Neo X Grind Hackathon:** 2nd place in India (∼200 teams)
- **Stellar Builders Camp:** 3rd place in India (80 solo developers)
- **Technoxian World Robotics Championship:** International semifinalist, FPV Drone Racing
- **INSPIRE Internship, Government of India:** Selected among the Top 1% of Class X Students

Leadership

Exun Clan (School Technology Club) — Aug 2022 – May 2026
President — *New Delhi, India*
- Raised $15,000+ for the club's international tech symposium, which drew 15,000+ international participants
- Led development of event platforms and internal tooling; hosted 3 competitive programming contests on CodeChef
- Launched the Girls in Tech initiative, running programming and technology workshops for 200 students

Technical Skills

Languages: Python, Go, Java, JavaScript, TypeScript
ML/AI: PyTorch, Mask R-CNN, Computer Vision, RAG, LLM Pipelines
Backend/Infra: Flask, Quart, Node.js, SQLite, MongoDB, AWS EC2, Load Balancing
Frontend: React, Next.js
Tools: Git, AI-assisted development (Claude Code)